JONAN JOHNSON

jonan143ph@gmail.com | +91 77387 85153 | linkedin.com/in/jonan-johnson | github.com/jonan19 | jonan404.github.io

EDUCATION

Bachelor's Degree in Artificial Intelligence and Data Science *2023 - 2027*
K.J. Somaiya School of Engineering | CGPA: 9.17

Honors in Cybersecurity (additional credential) *2024 - 2027*
K.J. Somaiya School of Engineering | CGPA: 9.75

Kamaladevi Junior College *2020 - 2022*
Percentage: 67.83%

Arya Gurukul *2019 - 2020*
Percentage: 84.20%

SKILLS

Frameworks & Tools: PyTorch, LangChain, LangGraph, FAISS, FastAPI, CrewAI, Streamlit, Docker, Git, GitHub, GitHub Actions

Programming & Data: Python, NumPy, Pandas, scikit-learn, SQL, MySQL, MongoDB, REST APIs, Power BI, Excel

Domain Expertise: Machine Learning (Supervised & Unsupervised), Deep Learning, Generative AI, LLMs, RAG & Vector Search, Agentic AI & Multi-Agent Systems, Prompt Engineering, AI Evaluation (Accuracy, Relevance, Latency, Safety), NLP

WORK EXPERIENCE

Artificial Intelligence Intern - AnixAI *Aug 2025 - Nov 2025*

- Built a **modular multi-agent Generative AI system** (CrewAI + LangChain) with custom LLM-based Test Planning and Test Case Generator agents, orchestrated through a Python pipeline.
- Generated test strategies, coverage matrices, risk assessments, and test cases across multiple categories for telecom QA.
- Tested outputs and tracked quality metrics, applying varied prompting techniques to improve results at lower LLM cost.

Machine Learning Intern - KIAAR *Jan 2025 - Jan 2026*

- Developed an ML-driven pipeline to forecast soil salinity across **10,000 acres**, supporting agricultural planning for farmers and agri-scientists.
- Integrated environmental and remote-sensing data into a high-resolution spatial model, collaborating across research institutions to move it toward a field-ready system.

PROJECTS

Trust Before Text: A Verification-Gated RAG System for Reliable Question Answering ***Flagship*** *| Ongoing |* GitHub

- Built a **verification-first RAG system** (Python, Qdrant hybrid retrieval, DeBERTa NLI) whose LLM-free gate decides to answer or abstain before any model call.
- Made the decision injection-proof by design: **0 of 45 prompt-injection attacks** changed it, and planted evidence was blocked on **48 of 48 probes**.
- Evaluated it on two external benchmarks with preregistered tests and traced every number in the co-authored paper to a result file.

Customer Churn Prediction & LLM Cost-Accuracy Benchmark GitHub

- Engineered a **churn-prediction classifier** (Random Forest, XGBoost) on **7,000+ telecom accounts** with leak-proof preprocessing pipelines and class-imbalance handling, achieving **84% ROC-AUC** and catching **79% of at-risk customers**.
- Benchmarked zero-shot and few-shot LLM prompting (Groq gpt-oss-20b) against the classical model on identical data, uncovering a numeric-anchoring bias in few-shot outputs and a hidden reasoning-token cost unique to reasoning models.
- Quantified the cost-accuracy-latency tradeoff between classical ML and LLMs for structured prediction, proving the classical model wins on every ranking metric at **1/1000th the cost and 11x lower latency** - turning an AI-hype question into an evidence-based recommendation.

Soulace: A Mental Wellness Assistant *SIH 2025 |* GitHub

- Designed and deployed a **multi-agent LLM architecture** (CrewAI + LangChain) with emotion-aware routing and context-sensitive response generation for a mental wellness web platform.
- Developed the agent orchestration layer - modular prompt and context engineering, memory management, and task-specific specialization - to improve conversational relevance and coherence.
- Added safety-aware conversational controls and response filtering to reduce hallucinations and keep the system's behavior responsible in front of real users.

CERTIFICATIONS

- Deep Learning Specialization (Sequence Models, CNNs, Hyperparameter Tuning & Optimization) - Coursera
- Retrieval Augmented Generation (RAG) - DeepLearning.AI
- Introduction to FastAPI and Backend Development Fundamentals - Packt
- GitHub Actions Masterclass: From Beginner to Advanced - Packt
- Google Cloud Fundamentals: Core Infrastructure - Google Cloud Skills Boost